

Notes: Ossa (AER, 2014)
Trade Wars and Trade Talks with Data

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1 Big picture

- **Question.** How large would noncooperative trade barriers be if major economies set tariffs strategically, and how large are the gains from international trade-policy cooperation?
- **Main objects.** The paper computes three tariff concepts: *optimal tariffs* chosen unilaterally without retaliation, *Nash tariffs* chosen in a worldwide trade war with retaliation, and *cooperative tariffs* chosen in efficient trade negotiations.
- **Unified motive for protection.** The framework nests three channels: traditional terms-of-trade manipulation, new-trade profit shifting, and political-economy protection of favored sectors.
- **Quantitative setting.** The application covers seven regions - Brazil, China, the European Union, India, Japan, the United States, and the Rest of the World - and 33 agricultural and manufacturing industries in 2007.
- **Data.** The core calibration uses GTAP 8 industry-level trade, production, and tariff data. The paper also uses UN Comtrade data to estimate demand elasticities and tariff data from Market Access Map and TRAINS to calibrate political-economy weights.
- **Computational approach.** The model is solved in changes using a Dekle-Eaton-Kortum style hat algebra. This lets the author compute counterfactual general equilibrium outcomes without separately estimating every productivity, trade-cost, and variety parameter.
- **Main result on unilateral optimal tariffs.** Optimal tariffs are very high. In the full version with lobbying, average optimal tariffs are about 62%, and the imposing government can gain while others lose.
- **Main result on trade wars.** Nash trade-war tariffs are also very high, averaging about 63% in the full version. But once everyone retaliates, all governments lose. The mean government welfare loss is about 2.9%.
- **Main result on trade talks.** Negotiations from Nash tariffs generate large gains; negotiations from factual tariffs generate smaller but positive gains; and free trade is close to the efficiency frontier.
- **Role of politics.** Political-economy forces matter for the distribution of tariffs across sectors and countries, but the headline averages are similar with and without lobbying.
- **Economic takeaway.** Trade agreements are valuable not because each country lacks a unilateral incentive to protect, but because each country's protection imposes externalities on others. The WTO/GATT logic is quantitatively important: retaliation turns individually attractive tariffs into collectively costly trade wars.

2 Incremental contribution of the paper

- **From qualitative theory to quantitative trade policy.** Theoretical trade-policy models had long emphasized terms-of-trade motives, profit-shifting motives, and lobbying motives. This paper puts them in one calibrated general equilibrium model and computes actual tariff counterfactuals.

- **A unified framework.** Earlier quantitative work usually isolated one motive: Armington/terms-of-trade effects, Krugman/new-trade relocation effects, or political economy. Ossa combines them and asks how much each matters in a realistic multi-country, multi-industry environment.
- **Disaggregated noncooperative tariffs.** The paper computes industry-level optimal and Nash tariffs for the major players in recent GATT/WTO negotiations, rather than imposing one aggregate tariff for each country pair.
- **Cooperative tariffs, not only counterfactual tariff cuts.** Many quantitative trade papers study exogenous tariff reductions. This paper solves for endogenous cooperative tariffs using a bargaining problem and compares negotiations starting from Nash, factual, and zero tariffs.
- **Computational contribution.** The analysis uses mathematical programming with equilibrium constraints to solve high-dimensional optimal, Nash, and cooperative tariff problems that would have been difficult with older algorithms.
- **Calibration contribution.** The model is calibrated directly to observed industry trade, production, and tariffs. The hat-algebra representation ensures that the factual economy is matched exactly at the level of observed trade flows and tariffs.
- **Political-economy calibration.** Political weights are inferred from factual tariffs. This gives a practical way to embed Grossman-Helpman-style lobbying in a quantitative general equilibrium model.
- **MFN and plurilateral negotiations.** The paper goes beyond unconstrained tariff policy and studies the Most Favored Nation principle. It shows that MFN barely changes global noncooperative tariff averages but strongly changes the incentives in smaller-group negotiations.
- **Policy contribution.** The paper provides magnitudes for three policy questions: how high trade-war tariffs would be, how costly a breakdown of cooperation would be, and how much scope remains for future multilateral liberalization.

3 Data and measurement

3.1 Countries, industries, and year

- The quantitative application focuses on the year 2007.
- The seven regions are Brazil, China, the European Union, India, Japan, the United States, and a residual Rest of the World.
- The 33 industries span agriculture and manufacturing. Examples include wheat, rice, dairy, wearing apparel, textiles, motor vehicles, machinery, electronics, chemicals, and wood products.
- The country set is chosen to capture the main players in recent GATT/WTO negotiations while keeping the computational problem manageable.

Interpretation: The model is neither a two-country toy model nor a fully global 200-country tariff game. It is a middle ground: rich enough to capture major strategic interactions, but compact enough to solve high-dimensional tariff problems.

3.2 Trade, production, and tariff data

- The main data source is GTAP 8, which provides industry-level trade flows, production, and tariffs.
- Trade flows are expressed at world prices, and tariffs are introduced as ad valorem wedges.
- The model requires industry expenditure shares, bilateral trade flows, and tariff revenues.
- Because the static model does not naturally contain aggregate trade imbalances, the paper first purges the data of aggregate imbalances and then uses the balanced dataset for all counterfactuals.

Interpretation: Matching the observed data is crucial. The counterfactuals start from an economy that exactly replicates observed industry-level trade and tariffs, so the results are not driven by a poor factual fit.

3.3 Demand elasticities

- The model needs an elasticity of substitution σ_s for each industry.
- Ossa estimates these elasticities using the Feenstra method applied to UN Comtrade data from 1994 to 2008.
- The estimation pools across the main importers in the analysis, so σ_s varies by industry but not by importing country.
- Estimated elasticities range from very high for relatively homogeneous goods, such as wheat and rice, to lower values for differentiated industries.
- The mean estimated elasticity is about 3.42.

Interpretation: The elasticity estimates are central. Low elasticities imply more market power and stronger terms-of-trade/profit-shifting motives, which push noncooperative tariffs up. The paper therefore reports extensive sensitivity checks that rescale σ_s .

3.4 Political-economy weights

- Government objectives attach weight λ_{js} to industry- s welfare in country j .
- These weights are scaled so that the average λ_{js} within a country equals one.
- A weight above one means the government values income in that industry more than income in an average industry.
- The weights are calibrated from observed tariffs using Market Access Map and TRAINS tariff data.
- Table 1 highlights the five highest weights for each country, showing which industries appear politically favored.

Interpretation: The political weights should not be read as direct measures of bribes or campaign contributions. They are reduced-form weights summarizing how much a government appears to care about industry-specific income when observed tariffs are rationalized through the model.

4 Model

4.1 Preferences and demand

Consumers in country j have nested CES/Cobb-Douglas preferences over industry varieties:

$$U_j = \prod_s \left(\sum_i \int_0^{M_{is}} x_{ijs}(\nu_{is})^{\frac{\sigma_s-1}{\sigma_s}} d\nu_{is} \right)^{\frac{\sigma_s-1}{\sigma_s} \mu_{js}}.$$

- x_{ijs} is consumption in country j of an industry- s variety produced in country i .
- M_{is} is the mass of varieties in industry s produced by country i .
- $\sigma_s > 1$ is the elasticity of substitution across varieties within industry s .
- μ_{js} is the expenditure share of country j on industry s .

Utility maximization implies demand

$$x_{ijs} = \frac{(p_{is} \theta_{ijs} \tau_{ijs})^{-\sigma_s}}{P_{js}^{1-\sigma_s}} \mu_{js} X_j,$$

where $\tau_{ijs} = 1 + t_{ijs}$ is one plus the ad valorem tariff.

Interpretation: Tariffs raise consumer prices for imports. The elasticity σ_s governs how strongly consumers substitute away from tariffed foreign varieties.

4.2 Firms, markups, and production

Each firm produces a differentiated variety under a simple inverse production function:

$$l_{is} = \sum_j \frac{\theta_{ijs} x_{ijs}}{\phi_{is}}.$$

Firms are monopolistic competitors and charge a constant markup:

$$p_{is} = \frac{\sigma_s}{\sigma_s - 1} \frac{w_i}{\phi_{is}}.$$

- θ_{ijs} is an iceberg trade barrier.
- ϕ_{is} is productivity.
- w_i is the wage in country i .
- Markups are higher in low-elasticity industries.

Interpretation: The model contains a new-trade profit-shifting channel because firms have positive markups and profits. Tariffs can reallocate production and profits toward domestic firms, especially in industries with high markups.

4.3 Government objective

Country j 's government maximizes

$$G_j = \sum_s \lambda_{js} W_{js},$$

where $W_{js} = X_{js}/P_j$ is industry-level real income.

- If $\lambda_{js} = 1$ for all industries, the government maximizes aggregate welfare.
- If some $\lambda_{js} > 1$, the government gives extra weight to politically influential industries.
- Industry income includes labor income, profits, and an allocated share of tariff revenue.

Interpretation: The political-economy extension changes which sectors governments want to protect. It can make high- λ sectors receive more protection even if pure welfare motives would not select them.

4.4 Equilibrium in levels

For given tariffs, equilibrium is characterized by profits π_{is} , wages w_i , price indices P_{js} , and total expenditure X_j . The four key conditions are: profits from markup pricing and demand, labor-market clearing, CES price-index aggregation, and the country budget constraint including tariff revenue and profits.

Interpretation: Solving the model directly in levels would require estimating many difficult objects: variety masses M_{is} , productivity ϕ_{is} , and trade barriers θ_{ijs} . The paper avoids this problem by solving the model in changes.

4.5 Hat algebra for counterfactuals

A hat denotes the ratio of a counterfactual value to its factual value. The key counterfactual system is

$$\begin{aligned} \hat{\pi}_{is} &= \sum_j \alpha_{ijs} (\hat{\tau}_{ijs})^{-\sigma_s} \left(\frac{\hat{w}_i}{\hat{P}_{js}} \right)^{1-\sigma_s} \hat{X}_j, \\ \hat{w}_i &= \sum_s \delta_{is} \hat{\pi}_{is}, \\ \hat{P}_{js} &= \left[\sum_i \gamma_{ijs} (\hat{w}_i \hat{\tau}_{ijs})^{1-\sigma_s} \right]^{\frac{1}{1-\sigma_s}}, \end{aligned}$$

with a corresponding budget constraint for $\hat{X}_j$.

- α_{ijs} , γ_{ijs} , and δ_{is} are observed-share terms constructed from factual trade flows, tariffs, profits, and expenditure.
- The unknown structural primitives drop out.
- The counterfactual tariff vector determines wage, expenditure, profit, and price-index changes.

Interpretation: This is the computational engine of the paper. It turns a difficult structural model into a system that can be solved using observables and elasticities.

5 Welfare channels

5.1 Approximate welfare decomposition

The paper uses a log-linear approximation to clarify the welfare channels:

$$\frac{\Delta W_j}{W_j} \approx \sum_{i,s} \frac{T_{ijs}}{X_j} \left(\frac{\Delta p_{js}}{p_{js}} - \frac{\Delta p_{is}}{p_{is}} \right) + \sum_s \frac{\pi_{js}}{X_j} \left(\frac{\Delta \pi_{js}}{\pi_{js}} - \frac{\Delta p_{js}}{p_{js}} \right) + \sum_{i,s} \frac{t_{ijs} T_{ijs}}{X_j} \left(\frac{\Delta T_{ijs}}{T_{ijs}} - \frac{\Delta p_{is}}{p_{is}} \right).$$

- The first term is the **terms-of-trade effect**: a country gains when its production prices rise relative to its consumption prices.
- The second term is the **profit-shifting effect**: a country gains when output and profits shift toward its high-markup industries.
- The third term is the **trade-volume/tariff-revenue effect**: tariff revenue changes with import volumes and prices.

Interpretation: A tariff is attractive unilaterally because it can improve the home terms of trade and shift profits toward domestic firms. But those gains are partly externalized losses on trading partners, which is why retaliation is destructive.

5.2 Example: a US tariff increase

The paper illustrates the channels with a counterfactual 50 percentage point increase in the US tariff on chemicals or apparel.

- Chemicals have a low elasticity and relatively high markups.
- Apparel has a high elasticity and relatively low markups.
- The US wage rises in both cases, generating a positive terms-of-trade effect.
- The profit-shifting effect is positive for chemicals but negative for apparel.

Interpretation: The same tariff increase can have different welfare effects depending on the protected industry's markup and substitution elasticity. This is why industry-level tariff structure matters.

6 Noncooperative and cooperative tariff concepts

6.1 Optimal tariffs

An optimal tariff vector is the tariff vector a country would choose if other countries kept their factual tariffs fixed and did not retaliate:

$$\max_{\{t_{ijs}\}_{i,s}} G_j \quad \text{subject to general equilibrium.}$$

- Optimal tariffs are unilateral best responses to factual foreign tariffs.
- They show how much a country would like to exploit foreign exporters if it could do so without retaliation.

- In the baseline without lobbying, they are driven by terms-of-trade and profit-shifting motives.
- With lobbying, high- λ industries receive extra protection.

Interpretation: Optimal tariffs are not a policy recommendation. They are a benchmark measuring the strength of unilateral incentives to protect.

6.2 Nash tariffs and trade wars

Nash tariffs are the mutually best responding tariff vectors:

$$t_j^N \in \arg \max_{t_j} G_j(t_j, t_{-j}^N) \quad \text{for each country } j.$$

- Each country chooses tariffs taking all other countries' tariff vectors as given.
- Every country tries to extract terms-of-trade and profit-shifting gains.
- In equilibrium, all countries impose high tariffs.
- Trade volumes fall sharply and every country loses.

Interpretation: The central externality is clear: each government's optimal tariff improves its own objective partly by reducing other governments' welfare. When everyone does this, the private gains disappear and the efficiency costs remain.

6.3 Cooperative tariffs

Cooperative tariffs are determined by a Nash bargaining protocol starting from different threat points:

- starting from Nash tariffs,
- starting from factual tariffs,
- starting from zero tariffs.

The negotiated tariffs lie on the efficiency frontier subject to feasibility restrictions. The paper usually rules out import subsidies by imposing a zero lower bound on tariffs.

Interpretation: The three starting points answer three different questions. Starting from Nash measures the full value of cooperation. Starting from factual tariffs measures remaining gains from future trade talks. Starting from zero tariffs asks whether free trade is close to efficient.

6.4 MFN principle

The Most Favored Nation principle restricts a country to charge the same tariff to all trading partners within an industry. The paper studies MFN versions of optimal, Nash, and cooperative tariffs.

- MFN has small effects on global optimal and Nash tariff averages because the unconstrained noncooperative tariffs vary much more across industries than across trading partners.
- MFN matters more for smaller coalitions of countries because it multilateralizes concessions: reducing tariffs for one partner also reduces them for others.
- This changes the incentives and payoff distribution in plurilateral negotiations.

Interpretation: MFN is not just a legal detail. It affects whether a subset of countries can bargain over bilateral liberalization without automatically extending benefits to nonparticipants.

7 Identification, calibration, and empirical design

7.1 What the design identifies

- The paper does not estimate a causal effect of a historical tariff change using a natural experiment.
- Instead, it uses a calibrated structural model to identify counterfactual tariff equilibria and welfare changes.
- The main identifying content comes from observed trade flows, observed tariffs, industry elasticities, and the structure of the model.
- The model is disciplined by exact matching of factual trade and tariff data.

Interpretation: The results should be read as model-based quantitative counterfactuals. Their credibility depends on whether the model captures the main strategic and general equilibrium channels of tariff policy.

7.2 Key assumptions

- Preferences are CES within industries and Cobb-Douglas across industries.
- Firms are monopolistically competitive with constant markups determined by σ_s .
- The number of varieties is exogenous in the main version.
- Labor is the only factor, so terms-of-trade effects operate through relative wages.
- Tariffs are the only policy instrument in the baseline analysis.
- The factual data are purged of aggregate trade imbalances before counterfactuals are computed.
- Political-economy weights are inferred from observed tariff patterns.

Interpretation: These assumptions make the model tractable and transparent, but also shape the magnitudes. For example, variable markups, input-output linkages, or multiple mobile factors could alter the composition of welfare effects.

7.3 Sensitivity strategy

- The paper varies the average elasticity of substitution to check whether the main results are driven by a particular elasticity estimate.
- Higher elasticities reduce market power and therefore reduce optimal and Nash tariff incentives.
- The results are reported for both baseline and political-economy versions.
- The paper also studies MFN restrictions and plurilateral liberalization exercises.

Interpretation: The qualitative messages are robust: noncooperative tariffs are high, trade wars are costly, and most cooperative gains have already been achieved relative to factual tariffs. The exact magnitudes fall as substitution elasticities increase.

8 Tables and figures

8.1 Original titles of extracted tables and figures

- **Table 1:** Parameter Estimates.
- **Table 2:** Effects of 50 Percentage Point Increase in US Tariff.
- **Figure 1:** Optimal Tariffs without Lobbying.
- **Table 3:** Optimal Tariffs.
- **Figure 2:** Optimal Tariffs with Lobbying.
- **Figure 3:** Nash Tariffs without Lobbying.
- **Table 4:** Nash Tariffs.
- **Figure 4:** Nash Tariffs with Lobbying.
- **Figure 5:** Cooperative Tariffs without Lobbying Starting at Nash Tariffs.
- **Figure 6:** Cooperative Tariffs without Lobbying Starting at Factual Tariffs.
- **Figure 7:** Cooperative Tariffs without Lobbying Starting at Zero Tariffs.
- **Table 5:** Cooperative Tariffs Starting at Nash Tariffs.
- **Table 6:** Cooperative Tariffs Starting at Factual Tariffs.
- **Table 7:** Cooperative Tariffs Starting at Zero Tariffs.
- **Table 8:** Optimal Tariffs under MFN.
- **Table 9:** Nash Tariffs under MFN.
- **Table 10:** Cooperative Tariffs under MFN Starting at Nash Tariffs.
- **Table 11:** Cooperative Tariffs under MFN Starting at Factual Tariffs.
- **Table 12:** Cooperative Tariffs under MFN Starting at Zero Tariffs.
- **Figure 8:** Discriminatory versus Nondiscriminatory Tariff Reductions.

The captions below use the original table and figure names from the paper so that each extracted visual is identifiable on its own.

8.2 Table 1: Parameter Estimates; Table 2: Effects of 50 Percentage Point Increase in US Tariff

Read:

- Table 1, *Parameter Estimates*, reports the estimated industry elasticities σ_s and country-industry political-economy weights λ_{js} .
- The highest elasticities are in goods such as wheat and rice; the lowest are in more differentiated categories such as other animal products and forestry.

- The political weights are scaled to have country means equal to one. Boxes mark the five highest weights for each country.
- Table 2, *Effects of 50 Percentage Point Increase in US Tariff*, illustrates how a 50 percentage point US tariff increase affects wages, production, welfare, terms of trade, and profit shifting.
- The chemical tariff raises US welfare slightly because it combines a positive terms-of-trade effect with a positive profit-shifting effect. The apparel tariff lowers US welfare because the profit-shifting effect is negative.

Economic message: Table 1 supplies the core calibration inputs. Table 2 explains why industry detail is essential: tariffs in low-elasticity/high-markup sectors have different welfare consequences from tariffs in high-elasticity/low-markup sectors.

TABLE 1—PARAMETER ESTIMATES

	σ_s	λ_{BRA}	λ_{CHN}	λ_{EU}	λ_{IND}	λ_{JPN}	λ_{ROW}	λ_{US}
Wheat	10.07	1.29	1.68	1.66	1.67	1.91	1.27	1.24
Rice	7.01	1.23	1.64	1.51	1.38	1.91	1.23	1.19
Dairy	5.89	1.24	1.29	1.56	1.29	1.46	1.25	1.34
Wearing apparel	5.39	1.29	1.46	1.22	1.13	1.03	1.22	1.48
Other metals	4.47	1.11	0.80	1.06	1.05	0.94	1.08	1.16
Vegetable oils, etc.	4.03	1.13	1.19	1.09	1.50	0.96	1.08	1.03
Bovine meat products	3.89	1.05	1.16	1.44	1.05	0.95	1.11	1.07
Leather products	3.67	1.18	1.13	1.06	0.97	1.17	1.08	1.22
Ferrous metals	3.67	1.10	0.73	1.04	1.04	0.90	1.03	1.09
Other manufactures	3.53	1.15	1.22	1.06	1.00	0.86	1.04	1.23
Other cereal grains	3.32	0.98	1.22	1.26	1.04	1.72	1.41	0.91
Oil seeds	3.21	0.97	1.01	0.87	1.20	1.79	1.20	1.04
Other meat products	3.20	1.03	1.09	1.12	1.04	0.91	1.11	0.96
Beverages and tobacco products	2.92	1.01	1.45	1.28	1.51	0.93	1.09	1.24
Bovine cattle, etc.	2.91	0.92	0.56	1.07	0.92	0.94	0.91	0.81
Textiles	2.87	1.04	1.19	0.96	0.91	0.82	0.99	1.26
Other transport equipment	2.84	0.91	0.76	0.94	0.74	0.73	0.90	1.03
Plant-based fibers	2.80	0.96	0.58	0.77	0.82	0.73	0.90	0.81
Wool, etc.	2.76	0.91	0.85	0.00	0.86	1.37	0.91	0.64
Motor vehicles, etc.	2.75	0.99	1.18	0.99	0.96	0.73	0.92	0.90
Metal products	2.70	1.01	0.96	0.94	0.84	0.75	0.91	1.06
Sugar	2.69	0.92	1.22	1.10	1.34	1.27	1.00	0.85
Other food products	2.62	0.90	1.01	0.98	1.05	0.93	0.96	0.89
Paper products, etc.	2.56	0.93	0.67	0.88	0.81	0.73	0.86	0.91
Other crops	2.53	0.92	0.64	0.87	1.12	0.74	0.93	0.80
Electronic equipment	2.49	0.91	0.82	0.87	0.61	0.66	0.84	1.00
Other mineral products	2.47	0.92	0.87	0.87	0.80	0.72	0.89	1.02
Other machinery, etc.	2.46	0.94	0.76	0.86	0.79	0.73	0.83	1.00
Vegetables, etc.	2.42	0.82	0.96	0.83	1.01	0.90	0.90	0.80
Chemical products, etc.	2.34	0.85	0.74	0.84	0.73	0.71	0.81	0.96
Wood products	2.32	0.90	0.77	0.79	0.72	0.71	0.83	0.90
Forestry	2.20	0.77	0.78	0.66	0.61	0.70	0.76	0.61
Other animal products	1.91	0.68	0.59	0.56	0.50	0.72	0.73	0.53
Mean	3.42	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Notes: The entries under " σ_s " are the estimated elasticities of substitution. The entries under " λ_{BRA} " until " λ_{US} " are the estimated political economy weights for each country which are scaled to have a mean of one. The boxes highlight the industries with the five highest values of λ for each country.

TABLE 2—EFFECTS OF 50 PERCENTAGE POINT INCREASE IN US TARIFF

	Δ US wage	Δ US production (protected)	Δ US production (other)
<i>Panel A. General equilibrium effects</i>			
Chemicals	1.45	5.73	-1.40
Apparel	0.67	33.35	-0.97
	Δ US welfare	Terms-of-trade effect	Profit shifting effect
<i>Panel B. Welfare effects</i>			
Chemicals	0.17	0.34	0.12
Apparel	-0.14	0.16	-0.15

Notes: The entries in panel A are the percentage change in the US wage normalized such that the average wage change across all countries is zero (column 1), the percentage change in the quantity of output in the US chemicals or apparel industry (column 2), and the average of the percentage changes in the quantity of output in the other US industries (column 3). The entries in panel B are the percentage change in US welfare (column 1), the component due to terms-of-trade effects (column 2), and the component due to profit shifting effects (column 3). The values in column 2 and 3 do not add up to the value in column 1 because they are computed using equation (14) which is a linear approximation. Here, all changes are computed relative to factual tariffs.

(a) Table 1: Parameter Estimates

(b) Table 2: Effects of 50 Percentage Point Increase in US Tariff

8.3 Figure 1: Optimal Tariffs without Lobbying; Table 3: Optimal Tariffs

Read:

- Figure 1, *Optimal Tariffs without Lobbying*, plots optimal tariffs without lobbying for each country and industry.
- Industries are ranked from lowest to highest elasticity. Optimal tariffs decline strongly with the elasticity of substitution.
- The median optimal tariffs are high for all countries, with little variation across trading partners relative to variation across industries.

- Table 3, *Optimal Tariffs*, reports welfare, wage, and profit effects of imposing optimal tariffs, both without and with lobbying.
- Without lobbying, the imposing country gains on average while other countries lose. With lobbying, government welfare gains are similar, though aggregate welfare and profit effects can differ.
- Panel C shows that higher elasticities reduce the welfare gains from unilateral tariffs.

Economic message: The optimal-tariff results quantify the temptation to defect from cooperation. A country can gain unilaterally by taxing foreign exporters, but those gains are externalized costs to trading partners.

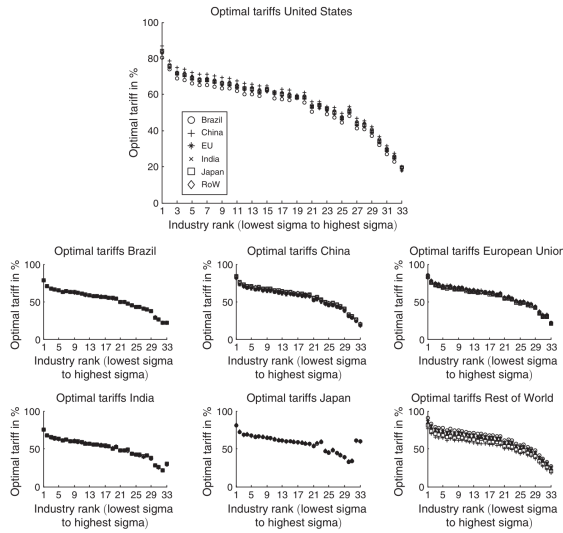


FIGURE 1. OPTIMAL TARIFFS WITHOUT LOBBYING

Note: The median optimal tariffs are 60.3 percent for the United States, 56.1 percent for Brazil, 59.3 percent for China, 61.3 percent for the European Union, 54.0 percent for India, 59.6 percent for Japan, and 61.5 percent for the Rest of the World.

TABLE 3—OPTIMAL TARIFFS

	Δ gvt. welfare		Δ welfare		Δ wage		Δ profits	
	Own	Other	Own	Other	Own	Other	Own	Other
<i>Panel A. Optimal tariffs without lobbying</i>								
Brazil	1.1	-0.1	1.1	-0.1	18.2	-3.0	0.8	-0.0
China	1.8	-0.6	1.8	-0.6	17.6	-2.9	0.5	-0.1
European Union	1.9	-1.0	1.9	-1.0	22.5	-3.7	0.1	-0.2
India	1.7	-0.1	1.7	-0.1	8.7	-1.5	2.7	-0.1
Japan	4.0	-0.3	4.0	-0.3	18.6	-3.1	1.7	-0.1
Rest of World	2.9	-1.7	2.9	-1.7	19.0	-3.2	1.1	-0.6
United States	2.3	-0.9	2.3	-0.9	23.8	-4.0	0.6	-0.1
Mean	2.2	-0.7	2.2	-0.7	18.3	-3.1	1.1	-0.2
<i>Panel B. Optimal tariffs with lobbying</i>								
Brazil	0.9	-0.1	1.0	-0.1	18.1	-3.0	0.3	-0.0
China	1.5	-0.4	1.5	-0.5	13.3	-2.2	0.1	-0.0
European Union	2.2	-1.2	1.7	-1.1	27.0	-4.5	-0.9	0.1
India	0.5	-0.0	0.7	-0.0	11.4	-1.9	0.6	-0.0
Japan	2.6	-0.4	1.0	-0.4	30.0	-5.0	-1.4	0.1
Rest of World	2.9	-1.7	2.6	-1.8	21.9	-3.7	-0.1	-0.2
United States	2.5	-0.9	2.1	-0.9	26.4	-4.4	-0.2	0.0
Mean	1.9	-0.7	1.5	-0.7	21.2	-3.5	-0.2	0.0
σ	Δ gvt. welfare		Δ welfare		Δ wage		Δ profits	
mean	Own	Other	Own	Other	Own	Other	Own	Other
<i>Panel C. Sensitivity of optimal tariffs w.r.t. σ_s</i>								
Without lobbying (all values are means)								
3.5	2.2	-0.6	2.2	-0.6	17.6	-2.9	1.1	-0.2
5.0	1.7	-0.4	1.7	-0.4	9.1	-1.5	1.1	-0.2
6.5	1.5	-0.2	1.5	-0.2	5.4	-0.9	1.1	-0.2
With lobbying (all values are means)								
3.5	1.8	-0.6	1.5	-0.6	20.2	-3.4	-0.2	0.0
5.0	1.2	-0.4	0.9	-0.4	10.5	-1.7	-0.2	0.0
6.5	1.1	-0.3	0.7	-0.3	6.5	-1.1	-0.2	0.0

Notes: The entries under "gvt. welfare" are the percentage changes in G , the entries under "welfare" are the percentage changes in W , the entries under "wage" are the percentage changes in w normalized such that the average wage change across all countries is zero, and the entries under "profits" are the percentage changes in total profits due to changes in industry output. The entries under "own" are the effects on the tariff imposing country while the entries under "other" are the averages of the effects on all other countries. The last rows of panels A and B report averages. Panel C reports only such averages.

(a) Figure 1: Optimal Tariffs without Lobbying

(b) Table 3: Optimal Tariffs

8.4 Figure 2: Optimal Tariffs with Lobbying; Figure 3: Nash Tariffs without Lobbying

Read:

- Figure 2, *Optimal Tariffs with Lobbying*, repeats the optimal-tariff exercise with calibrated lobbying weights.
- Political economy creates visible deviations from the smooth elasticity ranking because some high-weight industries receive extra protection.
- Figure 3, *Nash Tariffs without Lobbying*, reports Nash tariffs without lobbying.
- Nash tariffs look similar to optimal tariffs in level and industry ranking, but their welfare implications are very different because all countries retaliate.

- The visual similarity between Figures 1 and 3 helps explain why trade wars generate high tariffs close to unilateral optimal levels.

Economic message: Lobbying changes the cross-industry distribution of protection, but the broader noncooperative logic remains. Trade-war tariffs are high because every country has the same unilateral incentive to manipulate prices and shift profits.

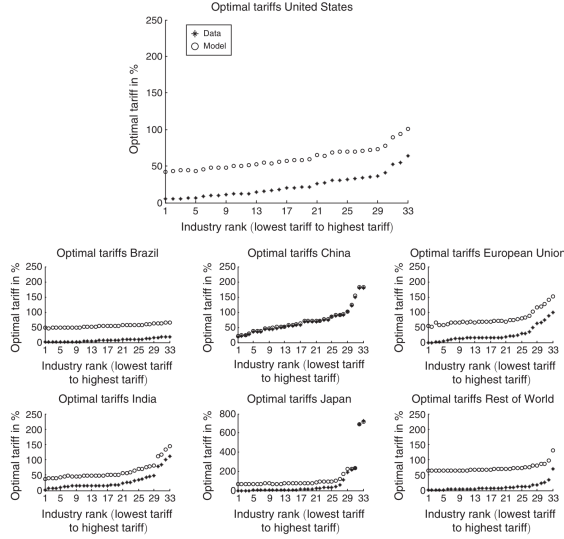


FIGURE 2. OPTIMAL TARIFFS WITH LOBBYING

Note: The median optimal tariffs are 56.4 percent for the United States, 54.2 percent for Brazil, 60.7 percent for China, 69.0 percent for the European Union, 49.9 percent for India, 77.5 percent for Japan, and 68.9 percent for the Rest of the World.

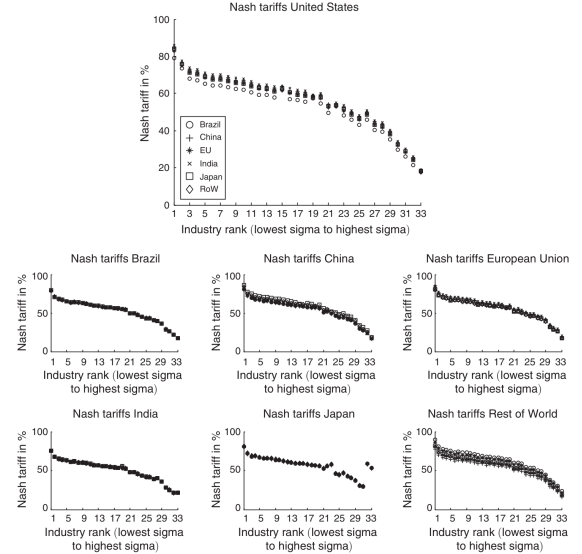


FIGURE 3. NASH TARIFFS WITHOUT LOBBYING

Note: The median Nash tariffs are 59.6 percent for the United States, 56.4 percent for Brazil, 58.6 percent for China, 59.1 percent for the European Union, 54.5 percent for India, 58.5 percent for Japan, and 59.7 percent for the Rest of the World.

(a) Figure 2: Optimal Tariffs with Lobbying

(b) Figure 3: Nash Tariffs without Lobbying

8.5 Table 4: Nash Tariffs; Figure 4: Nash Tariffs with Lobbying

Read:

- Table 4, *Nash Tariffs*, reports the welfare effects of Nash tariffs.
- In contrast to Table 3, there is only one multilateral tariff scenario: all countries choose their trade-war tariffs.
- Welfare falls for every country. In the full political-economy version, the mean government welfare loss is about 2.9%.
- Wage and profit effects vary by country, but the overall result is a global loss.
- Figure 4, *Nash Tariffs with Lobbying*, plots Nash tariffs with lobbying. The distribution is again high and broadly decreasing in the elasticity ranking, with political-economy deviations.

Economic message: The trade war result is the quantitative case for cooperation. The same instruments that look attractive in isolation become mutually destructive when used by all countries.

TABLE 4—NASH TARIFFS

	Δ gvt. welfare	Δ welfare	Δ wage	Δ profits
<i>Panel A. Nash tariffs without lobbying</i>				
Brazil	-1.9	-1.9	1.3	0.4
China	-2.2	-2.2	0.5	-0.2
European Union	-2.6	-2.6	2.7	-0.9
India	-2.2	-2.2	-9.3	1.9
Japan	-0.8	-0.8	-0.6	0.7
Rest of World	-5.0	-5.0	-0.8	-0.6
United States	-2.2	-2.2	6.3	-0.3
Mean	-2.4	-2.4	0.0	0.2

(Continued)

TABLE 4—NASH TARIFFS (Continued)

	Δ gvt. welfare	Δ welfare	Δ wage	Δ profits
<i>Panel B. Nash tariffs with lobbying</i>				
Brazil	-2.7	-2.5	-4.6	0.5
China	-3.4	-2.9	-7.1	0.3
European Union	-2.2	-2.7	5.6	-1.2
India	-3.6	-3.3	-10.5	0.8
Japan	-1.0	-2.8	11.4	-1.7
Rest of World	-5.3	-5.6	-1.3	-0.1
United States	-2.0	-2.4	6.5	-0.2
Mean	-2.9	-3.2	0.0	-0.2
σ_{mean}				
<i>Panel C. Sensitivity of Nash tariffs w.r.t. σ_s</i>				
Without lobbying (all values are means)				
3.5	-2.3	-2.3	0.0	0.2
5.0	-1.0	-1.0	0.0	0.3
6.5	-0.3	-0.3	0.0	0.2
With lobbying (all values are means)				
3.5	-2.8	-3.0	0.0	-0.2
5.0	-1.5	-1.7	0.0	-0.1
6.5	-0.8	-1.1	0.0	-0.1

Notes: The entries under "gvt. welfare" are the percentage changes in G , the entries under "welfare" are the percentage changes in W , the entries under "wage" are the percentage changes in w normalized such that the average wage change across all countries is zero, and the entries under "profits" are the percentage changes in total profits due to changes in industry output. The last rows of panels A and B report averages. Panel C reports only such averages.

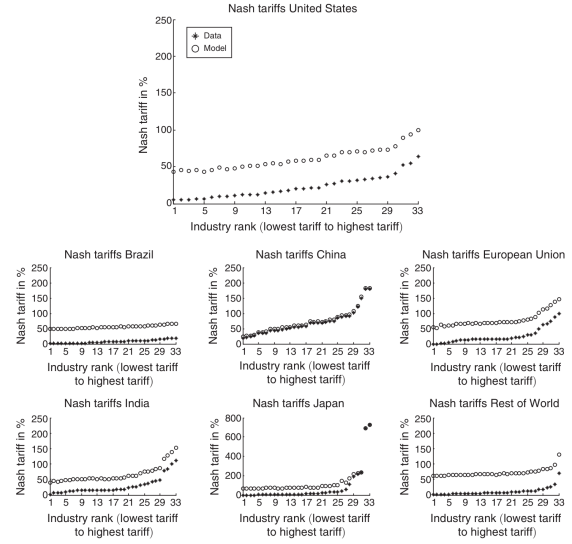


FIGURE 4. NASH TARIFFS WITH LOBBYING

Note: The median Nash tariffs are 56.6 percent for the United States, 54.7 percent for Brazil, 62.9 percent for China, 69.4 percent for the European Union, 54.1 percent for India, 77.6 percent for Japan, and 68.5 percent for the Rest of the World.

(a) Table 4: Nash Tariffs

(b) Figure 4: Nash Tariffs with Lobbying

8.6 Figure 5: Cooperative Tariffs without Lobbying Starting at Nash Tariffs; Figure 6: Cooperative Tariffs without Lobbying Starting at Factual Tariffs

Read:

- Figure 5, *Cooperative Tariffs without Lobbying Starting at Nash Tariffs*, shows cooperative tariffs without lobbying starting from Nash tariffs.
- Cooperative tariffs are much lower than Nash tariffs. Negotiation dismantles much of the mutually harmful protection created by the trade war.
- Figure 6, *Cooperative Tariffs without Lobbying Starting at Factual Tariffs*, shows cooperative tariffs starting from factual tariffs.
- The changes from factual tariffs are smaller because factual GATT/WTO tariffs are already far below the Nash trade-war benchmark.
- Some efficient tariff configurations would involve import subsidies, but the main analysis imposes a zero lower bound on tariffs.

Economic message: The contrast between Figures 5 and 6 is central. The total gains from cooperation are large relative to a trade war, but the remaining gains from today's factual tariffs are much smaller.

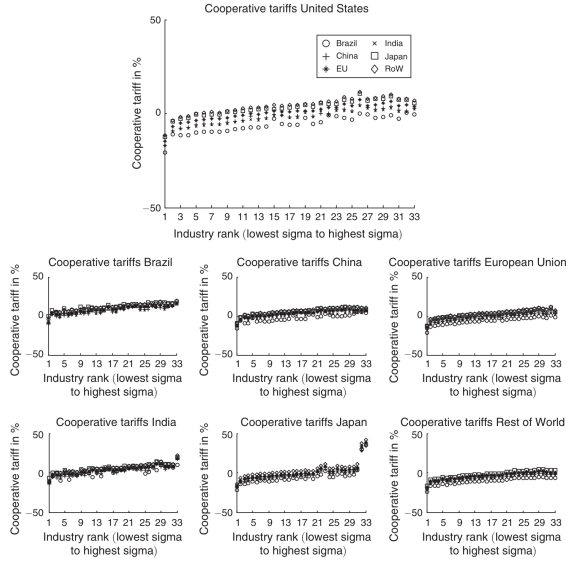


FIGURE 5. COOPERATIVE TARIFFS WITHOUT LOBBYING STARTING AT NASH TARIFFS

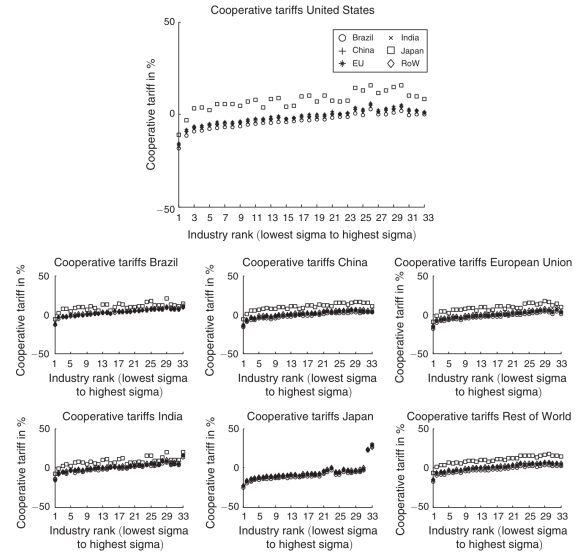


FIGURE 6. COOPERATIVE TARIFFS WITHOUT LOBBYING STARTING AT FACTUAL TARIFFS

(a) Figure 5: Cooperative Tariffs without Lobbying Starting at Nash Tariffs

(b) Figure 6: Cooperative Tariffs without Lobbying Starting at Factual Tariffs

8.7 Figure 7: Cooperative Tariffs without Lobbying Starting at Zero Tariffs; Table 5: Cooperative Tariffs Starting at Nash Tariffs

Read:

- Figure 7, *Cooperative Tariffs without Lobbying Starting at Zero Tariffs*, shows cooperative tariffs without lobbying starting from zero tariffs.
- The cooperative tariffs are close to zero, illustrating that free trade lies near the efficiency frontier.
- Table 5, *Cooperative Tariffs Starting at Nash Tariffs*, reports the welfare effects of cooperative negotiations starting at Nash tariffs.
- Without lobbying, all countries gain roughly 3.4%. With lobbying, average government welfare gains are about 3.6%.
- Wage and profit components differ across countries, but the aggregate welfare message is strongly positive.

Economic message: Negotiations starting from a trade war produce large gains because they remove the severe externalities built into Nash tariffs. This is the largest cooperative surplus considered in the paper.

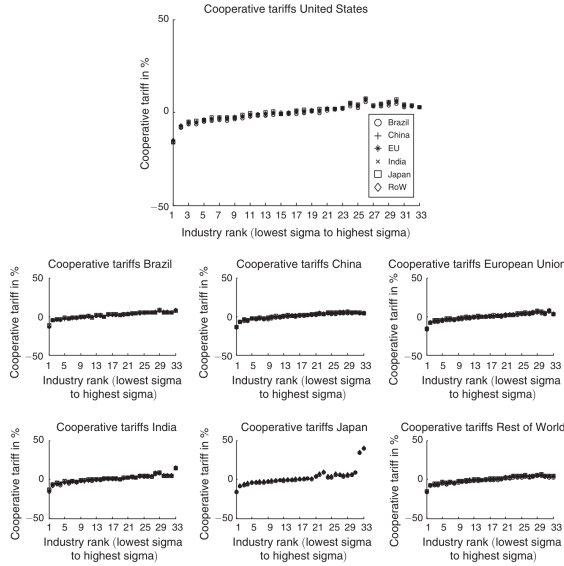


FIGURE 7. COOPERATIVE TARIFFS WITHOUT LOBBYING STARTING AT ZERO TARIFFS

TABLE 5—COOPERATIVE TARIFFS STARTING AT NASH TARIFFS

	Δ gvt. welfare	Δ welfare	Δ wage	Δ profits
<i>Panel A. Cooperative tariffs without lobbying</i>				
Brazil	3.4	3.4	9.2	-0.7
China	3.4	3.4	0.0	-0.8
European Union	3.4	3.4	-2.1	1.0
India	3.4	3.4	5.8	-0.9
Japan	3.4	3.4	-2.7	1.4
Rest of World	3.4	3.4	-6.0	0.6
United States	3.4	3.4	-4.2	0.4
Mean	3.4	3.4	0.0	0.1
<i>Panel B. Cooperative tariffs with lobbying</i>				
Brazil	3.6	3.5	10.7	-0.7
China	3.6	1.0	-4.7	-2.3
European Union	3.6	4.0	-2.7	1.4
India	3.6	3.6	5.7	-0.6
Japan	3.6	4.9	-0.8	1.7
Rest of World	3.6	4.2	-4.7	0.4
United States	3.6	4.1	-3.5	1.0
Mean	3.6	3.6	0.0	0.1
σ	Δ gvt. welfare	Δ welfare	Δ wage	Δ profits
<i>Panel C. Sensitivity of cooperative tariffs w.r.t. σ_s</i>				
Without lobbying (all values are means)				
3.5	3.3	3.3	0.0	0.1
5.0	2.2	2.2	0.0	-0.1
6.5	1.7	1.7	0.0	-0.2
With lobbying (all values are means)				
3.5	3.5	3.5	0.0	0.1
5.0	2.3	2.2	0.0	0.0
6.5	1.8	1.7	0.0	0.1

Notes: The entries under "gvt. welfare" are the percentage changes in G , the entries under "welfare" are the percentage changes in W , the entries under "wage" are the percentage changes in w normalized such that the average wage change across all countries is zero, and the entries under "profits" are the percentage changes in total profits due to changes in industry output. The last rows of panels A and B report averages. Panel C reports only such averages.

(a) Figure 7: Cooperative Tariffs without Lobbying Starting at Zero Tariffs

(b) Table 5: Cooperative Tariffs Starting at Nash Tariffs

8.8 Table 6: Cooperative Tariffs Starting at Factual Tariffs; Table 7: Cooperative Tariffs Starting at Zero Tariffs

Read:

- Table 6, *Cooperative Tariffs Starting at Factual Tariffs*, reports cooperative negotiations starting from factual tariffs.
- The average government welfare gain is much smaller than in Table 5: about 0.5% in the version with lobbying.
- Table 7, *Cooperative Tariffs Starting at Zero Tariffs*, reports cooperative negotiations starting from zero tariffs.
- The average government welfare gain from moving away from zero tariffs is tiny without lobbying and only about 0.2% with lobbying.
- These findings imply that factual tariffs are already much closer to the cooperative frontier than Nash tariffs are, and that free trade is close to efficient.

Economic message: Most of the potential cooperative surplus has already been captured by past trade negotiations. Future multilateral talks can still generate gains, but they are much smaller than the gains from preventing a reversion to trade-war policies.

	Δ gvt. welfare	Δ welfare	Δ wage	Δ profits
<i>Panel A. Cooperative tariffs without lobbying</i>				
Brazil	0.5	0.5	6.1	-0.7
China	0.5	0.5	0.2	-0.9
European Union	0.5	0.5	2.7	0.3
India	0.5	0.5	-4.0	1.0
Japan	0.5	0.5	-9.4	1.8
Rest of World	0.5	0.5	1.8	-0.2
United States	0.5	0.5	2.8	0.3
Mean	0.5	0.5	0.0	0.2
<i>Panel B. Cooperative tariffs with lobbying</i>				
Brazil	0.5	0.5	3.3	-0.1
China	0.5	-1.6	-8.0	-2.1
European Union	0.5	0.3	0.9	0.0
India	0.5	0.8	0.6	0.2
Japan	0.5	0.5	1.5	0.5
Rest of World	0.5	0.7	1.1	0.3
United States	0.5	0.6	0.6	1.0
Mean	0.5	0.3	0.0	0.0
σ	Δ gvt. welfare	Δ welfare	Δ wage	Δ profits
<i>Panel C. Sensitivity of cooperative tariffs w.r.t. σ_s</i>				
Without lobbying (all values are means)				
3.5	0.5	0.5	0.0	0.2
5.0	0.8	0.8	0.0	0.1
6.5	1.1	1.1	0.0	0.0
With lobbying (all values are means)				
3.5	0.5	0.3	0.0	0.0
5.0	0.7	0.4	0.0	0.0
6.5	1.0	0.6	0.0	0.0

Notes: The entries under "gvt. welfare" are the percentage changes in G , the entries under "welfare" are the percentage changes in W , the entries under "wage" are the percentage changes in w normalized such that the average wage change across all countries is zero, and the entries under "profits" are the percentage changes in total profits due to changes in industry output. The last rows of panels A and B report averages. Panel C reports only such averages.

	Δ gvt. welfare	Δ welfare	Δ wage	Δ profits
<i>Panel A. Cooperative tariffs without lobbying</i>				
Brazil	0.03	0.03	0.1	0.0
China	0.03	0.03	-0.2	0.2
European Union	0.03	0.03	0.1	0.0
India	0.03	0.03	-0.1	0.2
Japan	0.03	0.03	0.6	-0.2
Rest of World	0.03	0.03	-0.2	0.3
United States	0.03	0.03	-0.3	0.2
Mean	0.03	0.03	0.0	0.1
<i>Panel B. Cooperative tariffs with lobbying</i>				
Brazil	0.2	0.28	1.4	0.6
China	0.2	-1.25	-3.1	-1.4
European Union	0.2	-0.01	0.8	-0.1
India	0.2	-0.86	-0.7	-0.7
Japan	0.2	-0.44	1.0	-0.5
Rest of World	0.2	0.28	1.0	0.5
United States	0.2	0.15	-0.4	1.0
Mean	0.2	-0.27	0.0	-0.1
σ	Δ gvt. welfare	Δ welfare	Δ wage	Δ profits
<i>Panel C. Sensitivity of cooperative tariffs w.r.t. σ_s</i>				
Without lobbying (all values are means)				
3.5	0.03	0.03	0.0	0.1
5.0	0.01	0.01	0.0	0.1
6.5	0.01	0.01	0.0	0.1
With lobbying (all values are means)				
3.5	0.2	-0.26	0.0	-0.1
5.0	0.3	-0.29	0.0	0.1
6.5	0.4	-0.33	0.0	0.2

Notes: The entries under "gvt. welfare" are the percentage changes in G , the entries under "welfare" are the percentage changes in W , the entries under "wage" are the percentage changes in w normalized such that the average wage change across all countries is zero, and the entries under "profits" are the percentage changes in total profits due to changes in industry output. The last rows of panels A and B report averages. Panel C reports only such averages.

(a) Table 6: Cooperative Tariffs Starting at Factual Tariffs

(b) Table 7: Cooperative Tariffs Starting at Zero Tariffs

8.9 Table 8: Optimal Tariffs under MFN; Table 9: Nash Tariffs under MFN

Read:

- Table 8, *Optimal Tariffs under MFN*, repeats the optimal-tariff exercise under the MFN constraint.
- The results are almost identical to the unconstrained optimal-tariff results because unconstrained optimal tariffs vary mostly by industry, not by trading partner.
- Table 9, *Nash Tariffs under MFN*, repeats the Nash trade-war exercise under MFN.
- Again, the results are similar to the non-MFN case. MFN does not eliminate the central trade-war externality.
- Panel C in both tables confirms that the magnitudes fall as the assumed average elasticity rises.

Economic message: MFN does not by itself solve the noncooperative problem. It can restrict discrimination, but countries can still set high industry-level tariffs that harm trading partners.

TABLE 8—OPTIMAL TARIFFS UNDER MFN

	Δ gvt. welfare		Δ welfare		Δ wage		Δ profits	
	own	other	own	other	own	other	own	other
<i>Panel A. Optimal tariffs without lobbying under MFN</i>								
Brazil	1.1	-0.1	1.1	-0.1	18.2	-3.0	0.8	-0.0
China	1.8	-0.6	1.8	-0.6	17.6	-2.9	0.5	-0.1
European Union	1.9	-1.0	1.9	-1.0	22.4	-3.7	0.1	-0.2
India	1.7	-0.1	1.7	-0.1	8.7	-1.5	2.7	-0.1
Japan	4.0	-0.3	4.0	-0.3	18.6	-3.1	1.7	-0.1
Rest of World	2.8	-1.7	2.8	-1.7	19.0	-3.2	1.1	-0.6
United States	2.3	-0.9	2.3	-0.9	23.8	-4.0	0.6	-0.1
Mean	2.2	-0.7	2.2	-0.7	18.3	-3.1	1.1	-0.2
<i>Panel B. Optimal tariffs with lobbying under MFN</i>								
Brazil	0.9	-0.1	1.0	-0.1	18.1	-3.0	0.3	-0.0
China	1.5	-0.4	1.5	-0.5	13.3	-2.2	0.1	-0.0
European Union	2.2	-1.2	1.7	-1.1	27.0	-4.5	-0.9	0.1
India	0.5	-0.0	0.7	-0.0	11.4	-1.9	0.6	-0.0
Japan	2.6	-0.4	1.0	-0.4	30.0	-5.0	-1.4	0.1
Rest of World	2.9	-1.7	2.5	-1.8	22.0	-3.7	-0.1	-0.2
United States	2.5	-0.9	2.1	-0.9	26.4	-4.4	-0.2	0.0
Mean	1.9	-0.7	1.5	-0.7	21.2	-3.5	-0.2	0.0
σ	Δ gvt. welfare		Δ welfare		Δ wage		Δ profits	
mean	own	other	own	other	own	other	own	other
<i>Panel C. Sensitivity of optimal tariffs w.r.t. σ_s under MFN</i>								
Without lobbying (all values are means)								
3.5	2.2	-0.6	2.2	-0.6	17.6	-2.9	1.1	-0.2
5.0	1.7	-0.4	1.7	-0.4	9.0	-1.5	1.1	-0.2
6.5	1.5	-0.2	1.5	-0.2	5.4	-0.9	1.1	-0.2
With lobbying (all values are means)								
3.5	1.8	-0.6	1.5	-0.6	20.3	-3.4	-0.2	0.0
5.0	1.2	-0.4	0.9	-0.4	10.5	-1.8	-0.2	0.0
6.5	1.1	-0.3	0.7	-0.3	6.5	-1.1	-0.2	0.0

Notes: The entries under "gvt. welfare" are the percentage changes in G , the entries under "welfare" are the percentage changes in W , the entries under "wage" are the percentage changes in w normalized such that the average wage change across all countries is zero, and the entries under "profits" are the percentage changes in total profits due to changes in industry output. The entries under "own" are the effects on the tariff imposing country while the entries under "other" are the averages of the effects on all other countries. The last rows of panels A and B report averages. Panel C reports only such averages.

TABLE 9—NASH TARIFFS UNDER MFN

	Δ gvt. welfare	Δ welfare	Δ wage	Δ profits
<i>Panel A. Nash tariffs without lobbying under MFN</i>				
Brazil	-1.8	-1.8	2.0	0.5
China	-2.3	-2.3	0.0	-0.2
European Union	-2.5	-2.5	3.1	-0.9
India	-2.1	-2.1	-9.3	1.9
Japan	-0.7	-0.7	-0.7	0.8
Rest of World	-5.0	-5.0	-1.0	-0.6
United States	-2.2	-2.2	5.8	-0.3
Mean	-2.4	-2.4	0.0	0.2
<i>Panel B. Nash tariffs with lobbying under MFN</i>				
Brazil	-2.5	-2.4	-3.7	0.5
China	-3.5	-2.9	-7.4	0.3
European Union	-2.1	-2.6	6.1	-1.2
India	-3.6	-3.3	-10.5	0.8
Japan	-1.1	-2.9	10.7	-1.7
Rest of World	-5.3	-5.6	-1.4	-0.1
United States	-2.1	-2.5	6.1	-0.2
Mean	-2.9	-3.2	0.0	-0.2
σ_{mean}	Δ gvt. welfare	Δ welfare	Δ wage	Δ profits
<i>Panel C. Sensitivity of Nash tariffs w.r.t. σ_s under MFN</i>				
Without lobbying (all values are means)				
3.5	-2.3	-2.3	0.0	0.2
5.0	-1.0	-1.0	0.0	0.3
6.5	-0.2	-0.2	0.0	0.2
With lobbying (all values are means)				
3.5	-2.8	-3.0	0.0	-0.2
5.0	-1.5	-1.7	0.0	-0.1
6.5	-0.8	-1.1	0.0	-0.1

Notes: The entries under "gvt. welfare" are the percentage changes in G , the entries under "welfare" are the percentage changes in W , the entries under "wage" are the percentage changes in w normalized such that the average wage change across all countries is zero, and the entries under "profits" are the percentage changes in total profits due to changes in industry output. The last rows of panels A and B report averages. Panel C reports only such averages.

consistent with the fact that the above-mentioned "autonomous rate" "general rate"

(a) Table 8: Optimal Tariffs under MFN

(b) Table 9: Nash Tariffs under MFN

8.10 Table 10: Cooperative Tariffs under MFN Starting at Nash Tariffs; Table 11: Cooperative Tariffs under MFN Starting at Factual Tariffs

Read:

- Table 10, *Cooperative Tariffs under MFN Starting at Nash Tariffs*, reports cooperative tariffs under MFN starting from Nash tariffs.
- The welfare gains remain large because moving away from a trade war is valuable even when tariffs must be nondiscriminatory.
- Table 11, *Cooperative Tariffs under MFN Starting at Factual Tariffs*, reports cooperative tariffs under MFN starting from factual tariffs.
- The gains are smaller than from Nash tariffs, as in the unconstrained case.
- MFN affects the distribution of bargaining gains because tariff cuts granted to one partner must be extended to all partners.

Economic message: MFN changes the strategic nature of negotiations more than it changes global noncooperative tariff levels. It turns bilateral concessions into broader liberalization, which can dilute the incentive for small groups to negotiate.

TABLE 10—COOPERATIVE TARIFFS UNDER MFN STARTING AT NASH TARIFFS

	Δ gvt. welfare	Δ welfare	Δ wage	Δ profits
<i>Panel A. Cooperative tariffs without lobbying under MFN</i>				
Brazil	3.3	3.3	10.5	-0.4
China	3.3	3.3	-0.3	-0.5
European Union	3.3	3.3	-2.3	0.9
India	3.3	3.3	5.2	-0.4
Japan	3.3	3.3	-2.9	1.2
Rest of World	3.3	3.3	-6.0	0.3
United States	3.3	3.3	-4.2	0.3
Mean	3.3	3.3	0.0	0.2
<i>Panel B. Cooperative tariffs with lobbying under MFN</i>				
Brazil	3.4	3.3	10.2	-0.6
China	3.4	2.2	-1.0	-1.1
European Union	3.4	3.8	-3.6	1.2
India	3.4	3.4	4.2	-0.6
Japan	3.4	4.5	-0.7	1.3
Rest of World	3.4	4.0	-5.5	0.5
United States	3.4	3.8	-3.5	0.7
Mean	3.4	3.6	0.0	0.2
σ	Δ gvt. welfare	Δ welfare	Δ wage	Δ profits
<i>Panel C. Sensitivity of cooperative tariffs w.r.t. σ_x under MFN</i>				
Without lobbying (all values are means)				
3.5	3.2	3.2	0.0	0.2
5.0	2.1	2.1	0.0	0.0
6.5	1.6	1.6	0.0	-0.1
With lobbying (all values are means)				
3.5	3.3	3.5	0.0	0.2
5.0	2.0	2.1	0.0	0.1
6.5	1.5	1.7	0.0	0.1

Notes: The entries under "gvt. welfare" are the percentage changes in G , the entries under "welfare" are the percentage changes in W , the entries under "wage" are the percentage changes in w normalized such that the average wage change across all countries is zero, and the entries under "profits" are the percentage changes in total profits due to changes in industry output. The last rows of panels A and B report averages. Panel C reports only such averages.

TABLE 11—COOPERATIVE TARIFFS UNDER MFN STARTING AT FACTUAL TARIFFS

	Δ gvt. welfare	Δ welfare	Δ wage	Δ profits
<i>Panel A. Cooperative tariffs without lobbying under MFN</i>				
Brazil	0.3	0.3	3.3	-0.5
China	0.3	0.3	-0.6	-0.9
European Union	0.3	0.3	1.2	0.2
India	0.3	0.3	-5.9	1.0
Japan	0.3	0.3	0.6	0.5
Rest of World	0.3	0.3	0.5	-0.1
United States	0.3	0.3	0.8	0.5
Mean	0.3	0.3	0.0	0.1
<i>Panel B. Cooperative tariffs with lobbying under MFN</i>				
Brazil	0.3	0.3	2.5	-0.2
China	0.3	-0.5	-5.1	-1.1
European Union	0.3	0.2	0.2	0.0
India	0.3	0.5	0.9	0.2
Japan	0.3	0.3	0.4	0.4
Rest of World	0.3	0.5	0.4	0.2
United States	0.3	0.4	0.7	0.3
Mean	0.3	0.2	0.0	0.0
σ	Δ gvt. welfare	Δ welfare	Δ wage	Δ profits
<i>Panel C. Sensitivity of cooperative tariffs w.r.t. σ_x under MFN</i>				
Without lobbying (all values are means)				
3.5	0.3	0.3	0.0	0.1
5.0	0.6	0.6	0.0	0.0
6.5	0.8	0.8	0.0	-0.1
With lobbying (all values are means)				
3.5	0.3	0.3	0.0	0.0
5.0	0.4	0.3	0.0	-0.1
6.5	0.4	0.3	0.0	-0.2

Notes: The entries under "gvt. welfare" are the percentage changes in G , the entries under "welfare" are the percentage changes in W , the entries under "wage" are the percentage changes in w normalized such that the average wage change across all countries is zero, and the entries under "profits" are the percentage changes in total profits due to changes in industry output. The last rows of panels A and B report averages. Panel C reports only such averages.

(a) Table 10: Cooperative Tariffs under MFN Starting at Nash Tariffs

(b) Table 11: Cooperative Tariffs under MFN Starting at Factual Tariffs

8.11 Table 12: Cooperative Tariffs under MFN Starting at Zero Tariffs; Figure 8: Discriminatory versus Nondiscriminatory Tariff Reductions

Read:

- Table 12, *Cooperative Tariffs under MFN Starting at Zero Tariffs*, reports cooperative tariffs under MFN starting from zero tariffs.
- The gains are again very small, reinforcing the result that free trade is close to efficient.
- Figure 8, *Discriminatory versus Nondiscriminatory Tariff Reductions*, compares discriminatory and nondiscriminatory tariff reductions among the European Union, Japan, and the United States.
- Without MFN, liberalizing countries can gain more directly from tariff cuts among themselves.
- With MFN, tariff cuts also benefit nonparticipants. This reduces the payoff from limited-membership liberalization and can even make some participants worse off over parts of the tariff-cut range.

Economic message: Figure 8 shows why the institutional form of trade talks matters. MFN helps avoid discrimination, but it also means that nonparticipants receive some benefits of participants' concessions.

TABLE 12—COOPERATIVE TARIFFS UNDER MFN STARTING AT ZERO TARIFFS

	Δ gvt. welfare	Δ welfare	Δ wage	Δ profits
<i>Panel A. Cooperative tariffs without lobbying under MFN</i>				
Brazil	0.03	0.03	0.0	0.1
China	0.03	0.03	-0.2	0.2
European Union	0.03	0.03	0.1	0.0
India	0.03	0.03	-0.2	0.3
Japan	0.03	0.03	0.7	-0.1
Rest of World	0.03	0.03	-0.2	0.3
United States	0.03	0.03	-0.2	0.2
Mean	0.03	0.03	0.0	0.1
<i>Panel B. Cooperative tariffs with lobbying under MFN</i>				
Brazil	0.1	0.17	0.6	0.5
China	0.1	-0.24	-0.3	-0.2
European Union	0.1	-0.10	0.2	-0.2
India	0.1	-0.30	-0.2	-0.3
Japan	0.1	-0.41	-0.6	-0.4
Rest of World	0.1	0.07	0.2	0.3
United States	0.1	0.08	0.1	0.4
Mean	0.1	-0.11	0.0	0.0
σ				
<i>Panel C. Sensitivity of cooperative tariffs w.r.t. σ_s under MFN</i>				
Without lobbying (all values are means)				
3.5	0.03	0.03	0.0	0.1
5.0	0.01	0.01	0.0	0.1
6.5	0.01	0.01	0.0	0.1
With lobbying (all values are means)				
3.5	0.09	-0.11	0.0	0.0
5.0	0.14	-0.15	0.0	0.1
6.5	0.20	-0.20	0.0	0.1

Notes: The entries under “gvt. welfare” are the percentage changes in G , the entries under “welfare” are the percentage changes in W , the entries under “wage” are the percentage changes in w normalized such that the average wage change across all countries is zero, and the entries under “profits” are the percentage changes in total profits due to changes in industry output. The last rows of panels A and B report averages. Panel C reports only such averages.

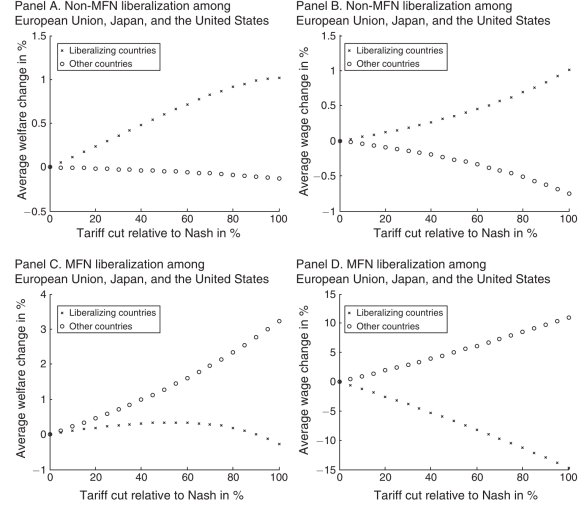


FIGURE 8. DISCRIMINATORY VERSUS NONDISCRIMINATORY TARIFF REDUCTIONS

(a) Table 12: Cooperative Tariffs under MFN Starting at Zero Tariffs

(b) Figure 8: Discriminatory versus Nondiscriminatory Tariff Reductions

9 Detailed result interpretation

9.1 Why optimal tariffs are high

- Large countries face upward-sloping foreign export supply curves in general equilibrium.
- A tariff can improve the home wage relative to foreign wages, raising the purchasing power of home production.
- Tariffs can also shift production toward domestic high-markup industries, increasing domestic profits.
- The incentive is strongest in industries with low substitution elasticities because markups and market power are larger.
- The large optimal tariffs should therefore be understood as the result of market power in trade, not as evidence that protection is globally efficient.

Interpretation: The paper’s high optimal tariffs are not surprising once trade policy is viewed as strategic. The surprise is that the fully general equilibrium, multi-industry model produces such large average tariffs even without explicitly assuming lobbying.

9.2 Why nobody wins the trade war

- If one country raises tariffs while others do not retaliate, it can gain at the expense of others.
- If all countries raise tariffs, each country’s terms-of-trade gains are offset by other countries’ retaliation.
- The result is lower trade volume, distorted production, and lower global real income.

- The political-economy version can make some industry groups better off, but government welfare falls on average.

Interpretation: A trade war is a prisoner’s dilemma. Tariffs are individually tempting but collectively harmful.

9.3 Why factual tariffs are much better than Nash tariffs

- Factual tariffs in 2007 are far below the Nash trade-war tariffs implied by the model.
- This gap is interpreted as evidence that past GATT/WTO negotiations achieved large cooperative gains.
- Moving from factual tariffs to the fully cooperative frontier still generates gains, but they are modest relative to the cost of abandoning cooperation.

Interpretation: The value of the WTO is largely defensive as well as progressive. Its biggest quantitative benefit may be preventing a return to noncooperative trade-war tariffs rather than delivering enormous additional gains from already-low factual tariffs.

9.4 Role of political economy

- Political weights tilt protection toward politically influential sectors.
- Some high-weight industries receive more protection than their elasticities alone would predict.
- However, political economy does not overturn the main average results: optimal and Nash tariffs remain high, trade wars remain costly, and cooperative gains from factual tariffs remain limited.

Interpretation: Lobbying matters for who is protected, but the strategic externality from tariffs is present even in a welfare-maximizing government model.

9.5 Role of elasticities

- Lower elasticities imply more market power, larger markups, and larger gains from manipulating trade prices.
- Higher elasticities make consumers substitute more easily and reduce the ability of tariffs to extract surplus.
- Sensitivity panels show that noncooperative tariff gains and trade-war losses fall as average elasticities rise.

Interpretation: The paper’s magnitudes are especially sensitive to elasticities, which are hard to estimate. The qualitative conclusions are robust, but precise welfare numbers should be interpreted with that uncertainty in mind.

10 Limitations and caveats

- **Model structure.** The model uses a one-factor framework with constant markups. Multiple factors, input-output linkages, or variable markups could change the channels.

- **Static environment.** The analysis is static. It does not directly model dynamic investment, innovation, firm entry, supply-chain formation, or long-run political adjustment.
- **Tariffs only.** The main policy instrument is the import tariff. Real-world trade policy also uses quotas, subsidies, product standards, procurement rules, antidumping duties, and non-tariff barriers.
- **Political weights.** The political-economy weights are calibrated from observed tariffs. They are reduced-form objects and may absorb factors other than lobbying.
- **Aggregate trade imbalances.** The paper purges trade imbalances before counterfactuals. This is practical, but it abstracts from macro-financial adjustment channels.
- **Country aggregation.** The Rest of the World is a large residual region. Its aggregation can mask strategic interactions among many smaller countries.
- **Interpretation of welfare.** Government welfare with lobbying is not the same as aggregate consumer welfare. Some results depend on whether one focuses on government objectives or aggregate welfare.

11 Short summary

- Ossa builds a multi-country, multi-industry quantitative trade-policy model that nests terms-of-trade, profit-shifting, and lobbying motives.
- The model is calibrated to 2007 trade and tariff data for seven regions and 33 industries.
- Optimal unilateral tariffs are high because countries can use tariffs to manipulate prices and shift profits.
- Nash trade-war tariffs are similarly high but reduce welfare for every country.
- Cooperative negotiations starting from Nash tariffs produce large gains; negotiations from factual tariffs produce smaller gains; and free trade is close to efficient.
- Political economy changes sectoral protection but not the overall message.
- MFN barely changes global noncooperative averages but matters for the incentives of smaller negotiating coalitions.

12 Conclusion

12.1 Core conclusion

The paper's core conclusion is that noncooperative trade policy incentives are quantitatively large. Countries would choose very high tariffs if unconstrained by trade agreements. However, when all countries act on those incentives, the result is a costly trade war in which every government is worse off.

12.2 Economic intuition

Tariffs allow a country to improve its terms of trade and shift profits toward domestic firms. These gains are attractive from the perspective of one country, but they come partly at the expense of others. Retaliation eliminates the unilateral gains while preserving the trade-volume and allocation distortions. This is why cooperation is valuable.

12.3 Incremental contribution revisited

The paper's distinctive contribution is to put modern quantitative trade methods to work on endogenous tariff choice. It moves the literature from asking, "What happens if tariffs fall by a specified amount?" to asking, "What tariffs would governments choose, and what would they negotiate?" That shift matters because optimal, Nash, and cooperative tariffs are endogenous objects shaped by the full general equilibrium environment.

12.4 Policy implication

The results support the view that international trade agreements solve a real strategic problem. They do not merely coordinate small technical tariff cuts; they prevent high-tariff equilibria that would be costly for all major players. At the same time, the small gains from additional talks starting at factual tariffs suggest that the largest historical gains from multilateral cooperation may already have been realized.

12.5 Main caveat

The analysis is powerful but model-dependent. Its welfare numbers rely on the model's elasticity estimates, its treatment of markups and political weights, and its static tariff-only environment. The best interpretation is therefore not that the exact numbers are immutable, but that the strategic trade-policy externality is large enough to matter quantitatively.

12.6 Takeaway

Trade wars are individually tempting and collectively destructive. Trade talks are valuable because they convert a high-tariff noncooperative equilibrium into a lower-tariff cooperative one. The key quantitative message is that the cost of losing trade-policy cooperation is much larger than the remaining gains from marginal additional liberalization at already-low factual tariffs.